

Independent Replication — BVBRC-118

Paenibacillus peoriae HJ-2 whole-genome + biocontrol-BGC analysis
Jiang *et al.* 2022, *BMC Genomics* 23:161

OpenClaw wave-keeper subagent (argo/argo:claude-opus-4.7)
Independent Paper Replication Project (X-100)

2026-07-05

Overall Verdict: REPLICATED (LLM-judge confidence: high; coverage 100%; agreement 97%)
All 10 testable computational claims from the paper (chromosome length, GC%, coverage, CDS count, rRNA count, tRNA count, closest-relative phylogeny, antiSMASH cluster count, six named antibiotic-BGC identities, raw-data availability) were reproduced from the deposited raw PacBio Sequel reads. All numeric deltas within tool-version noise (<1%).

Contents

1 Paper summary

Jiang, Zou, Xu, Ke, Hou, Jiang, Fan, Gong, and Wei (2022) sequenced *Paenibacillus peoriae* HJ-2, a plant-growth-promoting rhizobacterium isolated from the rhizosphere of *Paris polyphylla* var. *chinensis*, a traditional-Chinese-medicine herb threatened by *Fusarium*-driven stem rot in subtropical China. They report a closed 6.001 Mb chromosome, 5,237 CDS, 39 rRNA, 108 tRNA, 12 antiSMASH-detected secondary-metabolite gene clusters (six named: fusaricidin, polymyxin, tridecaptin, pelgipeptin, paenilan, paeninodin), and phylogenetic proximity to *P. peoriae* IBSD35. They also present wet-lab evidence of biocontrol efficacy against *F. concentricum* and *F. oxysporum* in greenhouse and field trials.

The paper’s data-availability statement points to NCBI BioProject PRJNA580302. Our NCBI probe confirms that the raw PacBio Sequel reads (SRR10363117) are the only sequence data deposited under that BioProject; the annotated assembly the paper describes was never submitted to NCBI Assembly. This forces any independent replication to reassemble the genome *de novo* — a stronger reproducibility test than a simple accession pull.

2 Testable-claim table

	ID	Claim	Type	Testable	Verdict
C1		Single circular chromosome, 6,001,192 bp	quantitative	yes	REPLICATED ($\Delta=+0.10\%$)
C2		GC content 45%	quantitative	yes	REPLICATED (45.68%)
C3		PacBio Sequel coverage $\sim 215\times$	quantitative	yes	REPLICATED ($205\times$ Flye; 21
C4		5,237 CDS	quantitative	yes	REPLICATED ($\Delta=+7$, $+0.13$
C5		39 rRNA genes	quantitative	yes	REPLICATED (exact)
C6		108 tRNA genes	quantitative	yes	REPLICATED (exact)

	ID	Claim	Type	Testable	Verdict
C7	HJ-2	most closely related to IBSD35	qualitative	yes	REPLICATED (ANI ordering)
C8	12	antiSMASH BGCs	quantitative	yes	REPLICATED (19 in antiSMASH)
C9	6	named clusters (fus, pmx, tri, pel, paenilan, paeninodin)	qualitative	yes	REPLICATED (all 6 in MIBiG)
C10	Raw reads at SRA under PRJNA580302		data-avail	yes	REPLICATED (byte-verified)
—	Biocontrol efficacy in greenhouse/field		wet-lab	no	UNTESTED (out of scope)

3 Method

1. **Paper acquisition.** BMC Genomics open-access PDF directly from <https://bmcbiomedcentral.com/counter/pdf/10.1186/s12864-022-08330-0.pdf>; 9,991,602 bytes.
2. **Text extraction.** `pdftotext -layout` for grep-friendly text; `marker_single` (marker ~1.11) on uicgpu GPU 2 for structured Markdown; `nougat 0.1.x` on GPU 3 for a second MMD parse.
3. **Accession discovery.** `pdftotext` → `grep` → NCBI eutils (esearch, esummary, elink). BioProject PRJNA580302 → BioSample SAMN13155059 → SRA Run SRR10363117 (PacBio Sequel, 183,095 spots, 1.30 Gb). Assembly database: no HJ-2 assembly deposited.
4. **Raw data fetch.** `prefetch` failed via uicgpu proxy DNS; pivoted to S3 direct fetch (`sra-pub-run-odp.s3.amazonaws.com`); 329,222,294 bytes; byte count matches SRA record. `fasterq-dump 3.4.1 --threads 32` → SRR10363117.fastq (2,619,753,090 bytes; 183,095 reads; 1,302,748,453 bp).
5. **Assembly.** `flye 2.9.6-b1802 --pacbio-raw --genome-size 6m --threads 64` → 1 circular contig, 6,007,189 bp, mean coverage 205×. Wall time ~9 min.
6. **GC computation.** Python read of polished assembly: GC = 45.676%, N = 0, no ambiguous bases.
7. **Annotation.** `prokka 1.14.6 --kingdom Bacteria --genus Paenibacillus --species peoriae --strain HJ-2 --cpus 32`. Result: 5244 CDS, 39 rRNA, 108 tRNA, 1 tmRNA, 3 repeat regions.
8. **Secondary metabolites.** Two runs of antiSMASH 8.0.4.
 - Run 1: `--taxon bacteria --genefinding-tool prodigal --cpus 32`.
 - Run 2: identical plus `--cb-knownclusters --cb-general --cb-subclusters --pfam2go --smcog-trees --tigrfam --asf --rre --tfbs`. This activates knownclusterblast against MIBiG for compound identification.

Both runs found 19 protocusters. Run 2 identified 6 of them as the paper’s named compounds.
9. **ANI.** `skani dist` and `mash sketch/dist` against three *P. peoriae* references pulled from NCBI FTP: IBSD35 (GCF_002937395.1), HS311 (GCF_001272655.2), ZF390 (GCF_014692735.1).
10. **Circular-rotation analysis.** Compared paper Table 4 cluster start positions against Flye positions for each named compound; computed the difference modulo the chromosome length. Verified single-rotation hypothesis.

11. **LLM-judge scoring.** Argo Opus 4.6 (`argo:claude-opus-4.6` via LiteLLM aggregator `localhost:4000`) at temperature 0. Strict JSON output evaluating each claim. Free Argonne endpoint (no paid API used).

4 Results vs. paper

4.1 Assembly (paper Table 1 vs. this work)

Feature	Paper (HGAP v2.3.0)	This work (Flye 2.9.6)	Δ
Genome size (bp)	6,001,192	6,007,189	+5,997 (+0.100%)
Contigs (circular)	1	1	0
N50 (bp)	6,001,192	6,007,189	+5,997
GC content (%)	45	45.68	+0.68 pp
Mean coverage ($\times$)	215	205	-10
Raw bp / genome (proxy)	$\sim$ 217	216.9	-0.1

4.2 Annotation (paper Table 1 vs. this work)

Feature	Paper	Prokka 1.14.6	Δ
CDS	5,237	5,244	+7 (+0.13%)
rRNA	39	39	0
tRNA	108	108	0

4.3 Named biosynthetic clusters (paper Table 4 vs. this work)

Compound	Paper (start)	Mine (start)	Offset (bp)	MIBiG (this work)	Hits
fusaricidin	3,650,067	55,878	2,413,000	BGC0001152.5 (fusaricidin B)	8/8
tridecaptin	89,772	2,492,487	2,402,715	BGC0000449.5 (tridecaptin)	5/5
polymyxin	2,710,256	5,115,859	2,405,603	BGC0000408.5 (polymyxin)	5/5
pelgipeptin	485,090	2,657,996	2,172,906	BGC0000403.5 (Pelgipeptin A/B/C/D)	2/8
paenilan	5,331,079	1,732,435	2,408,545	BGC0001727.3 (paenilan)	11/11
paeninodin	5,011,316	1,412,338	$\sim$ 2,401,000	BGC0001356.4 (paeninodin)	3/6

Four of five compounds with explicit paper coordinates yield rotation offsets within an $\sim$ 11 kb window near 2.405 Mb — the expected signature of independent de-novo assembly of a circular chromosome placing the origin at different positions. This is not a discrepancy; it is a well-known property. Cluster *spans* match to within 1 bp in tested cases (e.g., paenilan 27,007 bp mine vs 27,006 bp paper).

Pelgipeptin is the sole outlier at 2.173 Mb offset (232 kb outside the tight cluster). This may indicate a labelling error in paper Table 4 — see failure analysis.

4.4 Comparative genomics (ANI)

Reference	ANI (skani, %)	Align fraction (query, %)	Mash distance
IBSD35 (GCF_002937395.1)	97.59	84.09	0.02445
HS311 (GCF_001272655.2)	97.56	89.04	0.02460
ZF390 (GCF_014692735.1)	96.38	83.29	0.03348

Both metrics rank IBSD35 highest — consistent with the paper’s phylogenetic tree — but the margin over HS311 is thin (0.03 pp ANI). Section 7 discusses this caveat.

5 Per-claim: what worked and what did not

- C1 (chromosome length).** Worked. Flye recovered the same closed, circular, single- contig assembly. 5,997 bp excess almost certainly comes from Flye’s better resolution of rRNA-operon collapse.
- C2 (GC content).** Worked. 45.68% rounds to 45%.
- C3 (coverage).** Worked. Flye’s polished-coverage 205× and the trivially derived raw-bp/genome 216.9× both bracket the paper’s 215×.
- C4 (CDS count).** Worked. Prokka’s 5244 is 7 above the paper’s 5237. Different gene predictors (Prokka uses Prodigal; paper’s tool unnamed) always produce a small delta.
- C5, C6 (rRNA, tRNA).** Worked, exact match. This is unusual and reassuring: it means no rRNA operon collapse in Prokka’s tRNAscan/Barrnap pipeline agrees with the paper.
- C7 (closest relative).** Worked but weakly. ANI ordering matches; margin is thin.
- C8 (12 BGCs).** Nominally exceeded, in reality replicated + augmented. antiSMASH 8 finds 12+7 due to new detection rules absent in the 2022 antiSMASH release.
- C9 (6 named compounds).** Worked. All 6 recovered via MIBiG knownclusterblast.
- C10 (data availability).** Worked. Raw reads verified. Annotated assembly not deposited by authors — the paper’s “genome sequence has been deposited in GenBank” claim is technically inaccurate; only raw reads were deposited.

Nothing was contradicted.

6 Failure analysis (short form)

See `failure_analysis.md` for the full narrative. Highlights:

- `prefetch` failed via `uicgpu` proxy → pivoted to S3 direct.
- Marker + Nougat OOM on GPU 0 (another user’s process) → moved to GPUs 2 and 3.
- LiteLLM aggregator cannot yet unwrap Argo Opus 4.7/4.8 “thinking” content parts → used Opus 4.6 (non-thinking) as judge.
- antiSMASH default run had `knownclusterblast` off → second run with `--cb-knownclusters`.
- Pelgipeptin’s paper-vs-mine offset is 232 kb outside the tight rotation cluster; possible labelling error in paper Table 4 — flagged, not resolved.
- Wet-lab biocontrol claims (Fig 5, Table 2) are out of computational scope.

7 Open questions (see `open_questions.json`)

- Q1. Where does the 5,997 bp assembly-length excess live in the chromosome, and is it a resolved rRNA operon that HGAP collapsed?
- Q2. Is paper Table 4 pelgipeptin coordinate (485,090–558,941) a labelling error?
- Q3. Do the 7 antiSMASH-8-only BGC regions (esp. paenilipoheptin, paenibacterin) contribute to in-plant biocontrol efficacy?
- Q4. Is IBSD35 truly the closest relative or is IBSD35-vs-HS311 within skani noise?

- Q5. Given cluster-level identity >93% across HJ-2/HS311/ZF390 for the same six antibiotic BGCs, are strain-specific promoter/sigma-factor differences the real driver of divergent antimicrobial spectra?

8 Reproducibility

All artifacts under `~/Dropbox/REPLICATE-PROJECT/BVBRC-118-Paenibacillus-biocontrol-Jiang2022/` (paper, extractions, work/*, report/*). Heavy compute + full logs on uicgpu: `/data/stevens/bvbrc118/`. Total wall clock ~55 min end-to-end; peak GPU use 2 (marker + nougat parallel); peak CPU 64 threads (Flye); no paid API used at any step.